

Vision Belt: Intelligent Assistive Device for Collision Avoidance by Using Computer Vision Technology and Ultrasonic Sensors

Chi-Huang Hung

Graduate Institute of Applied Science
and Engineering,
Dept. of Information Technology
Fu Jen Catholic University,
Lee-Ming Institute of Technology
New Taipei City, Taiwan R.O.C
allen@mail.lit.edu.tw

Chun-Nan Chi

Graduate Institute of Applied Science
and Engineering Fu Jen Catholic
University
New Taipei City, Taiwan R.O.C
412068024@m365.fju.edu.tw

Shih-Yung Huang

Graduate Institute of Applied Science
and Engineering Fu Jen Catholic
University
New Taipei City, Taiwan R.O.C
409068037@m365.fju.edu.tw

Chia-Hsun Hsu

Dept. of International High School,
Kang Chiao International School
New Taipei City, Taiwan
ls11081@stu.kcislk.ntpc.edu.tw

Yi-An Lu

Dept. of Electrical Engineering
Lee-Ming Institute of Technology
New Taipei City, Taiwan R.O.C
lit00643@mail.lit.edu.tw

Abstract—Visually impaired individuals have been using white canes for walking on the road for many years. Although they are affordable and easy to use, there are still shortcomings in terms of providing sufficient reminders and warnings. While tactile paving on the floor provides a guide for visually impaired individuals to follow, they must walk on it with great concentration. The drawback is that sometimes it can lead to tripping or missing directional cues. White canes may not always offer sufficient assistance to those with limited abilities. Therefore, this study utilizes a miniature camera and embedded computer with computer vision technology to create a belt assistance system. This system helps visually impaired individuals walk on yellow tactile paving, and when combined with ultrasonic sensors, it extends the detection range from 2D to 3D. In this way, visually impaired individuals can understand whether there are obstacles ahead and if they are on the correct path through vibrations from left and right vibrating motors.

Keywords—Obstacle Avoidance, Visually Impaired Person, Auxiliary Equipment, Ultrasonic, Computer Vision Technology.

I. INTRODUCTION

For visually impaired individuals, safety during outdoor mobility is undoubtedly one of the most critical considerations. In busy or unfamiliar environments, the ability to timely recognize and avoid obstacles directly impacts their safety and autonomy. Research indicates that providing real-time and accurate environmental information is crucial for the outdoor mobility of visually impaired individuals [1]. Understanding the approximate locations of obstacles on the left and right sides, along with the relative distances to these obstacles, is crucial information. This not only helps them better plan their walking routes but also enhances their reaction time and obstacle avoidance capabilities when facing potential dangers.

In current research on assistive technologies for the visually impaired, many solutions attempt to utilize ultrasonic rangefinders and web cameras to detect obstacles in front of visually impaired individuals [2, 3]. These devices can provide information about obstacle distances and types, thereby assisting visually impaired individuals in navigating outdoor environments more safely. Furthermore, some studies integrate gyroscopes or GPS systems. When a fall is detected, these systems capture the current location and send emergency

notifications to specific individuals or organizations [4, 5], significantly improving the safety measures for visually impaired individuals in emergency situations.

This study proposes a visual assistance system for visually impaired individuals, integrating traditional ultrasonic sensors with intuitive feedback through computer vision technology. The embedded system, equipped with a camera, instantly monitors obstacles along the walking path, particularly on tactile paving. When potential obstacles are detected, the ultrasonic rangefinder precisely measures the distance. The system includes vibrating motors on the belt's sides, indicating obstacle orientation and proximity through varying frequencies. As obstacles approach, vibration frequency increases, providing continuous and accurate distance perception for timely obstacle avoidance. The prototype in Figure 1 demonstrates a visually impaired person using the system on a road with guide bricks. Integrating components into the belt head distributes weight, enhancing practicality and improving the quality of life for visually impaired individuals.

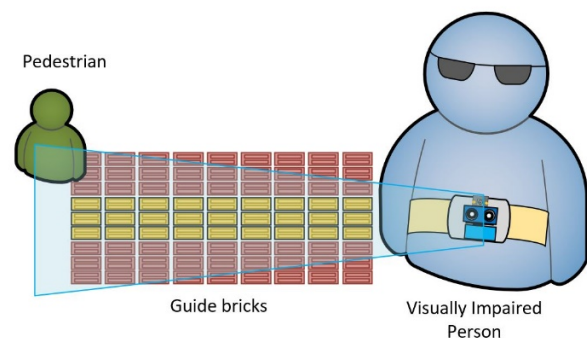


Fig. 1. shows a visually impaired person carrying the prototype of this study walking on a road paved with guide bricks, and there are passers-by in front of him.

II. HARDWARE ARCHITECTURE

The Vision Belt, installed on a belt, utilizes an embedded computer as its control center and is equipped with a camera, ultrasonic rangefinder, and vibration motors on the belt's sides (Figure 2). The embedded computer employs computer vision technology to differentiate between tactile paving and regular tiles. When visually impaired individuals deviate from the yellow tactile paving, a warning message is generated. The ultrasonic sensor, mounted on a servo motor for a 90-degree movement, enhances precision. Vibration motors convey obstacle orientation and distance, with increased frequency as obstacles approach.

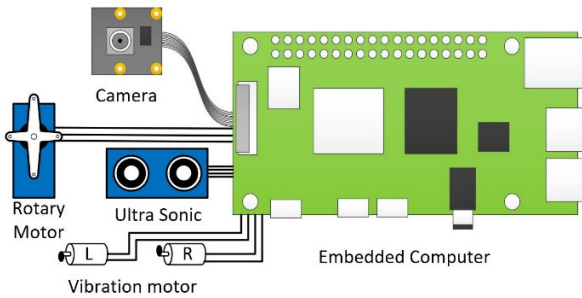


Fig. 2. Hardware architecture of Vision Belt.

III. SOFTWARE ARCHITECTURE

The software system runs on the Linux open-source operating system and is coded in Python. The workflow involves powering up the system, initiating the camera and ultrasonic rangefinder. The camera, using image processing, detects obstacles in the path of visually impaired individuals. If obstacles are detected, the ultrasonic rangefinder measures the relative distance. Based on the obstacle's orientation and proximity, the system activates the left or right vibration motor with varying frequencies, conveying information about the obstacle's approximate orientation and distance to visually impaired individuals. Figure 3 depicts the software flowchart.

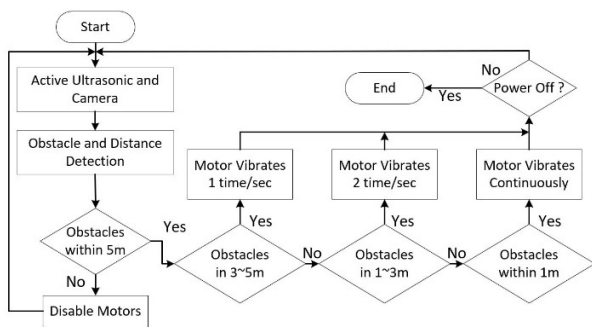


Fig. 3. Software flow chart of Vision Belt.

The obstacle detection algorithm processes camera-captured images with preprocessing steps, including grayscale conversion, Gaussian Blur, and Binarization, to eliminate unnecessary signals. The algorithm employs the Canny edge detection and Hough Transform to identify edges along the tactile paving as visually impaired individuals walk. It then identifies bounding boxes for obstacles, checking for intersections with the path boundaries. Upon detection, the ultrasonic sensor reads the obstacle's orientation and distance, triggering the corresponding vibration motor to alert the visually impaired individual.

IV. IMPLEMENTATION RESULTS

In Figure 4(a), the original camera-captured image is depicted. Figure 4(b) shows the extracted edges of the yellow floor tiles, establishing the walking path using HSV cylindrical-coordinate representations conversion. Subsequently, after employing edge detection algorithms and the YOLO algorithm, the resulting image in Figure 4(c) is obtained. When obstacles intersect the path of visually impaired individuals, our algorithm activates the ultrasonic sensor to measure obstacle distance. Vibration motors then convey obstacle proximity by varying vibration frequencies to visually impaired individuals."

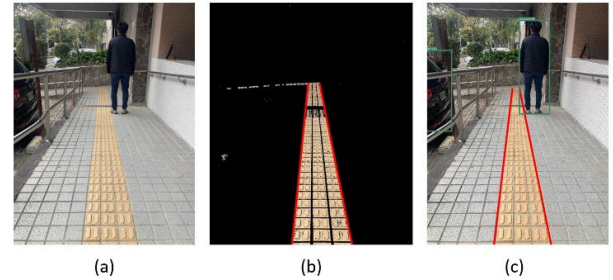


Fig. 4. (a) Original image, (b) edge drawing after HSV conversion, (c) image after processing.

CONCLUSION

In the field of assistive technologies for the visually impaired, methods using cameras, ultrasonic rangefinders, and vibration motors for obstacle navigation have been explored. However, our study demonstrates significant advantages in providing continuous and accurate obstacle orientation and distance information. Focusing on innovation, our system integrates advanced image processing and ultrasonic range finding. By using varying vibration frequencies, the system continually informs visually impaired individuals about obstacle orientation and distance, enhancing their independence and security in unfamiliar environments. Through ongoing research and development, we aim to provide substantial assistance to the visually impaired community, enabling them to explore the world with increased confidence and safety.

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